A Critical Reflection on Cloud Service Models: Architectural Boundaries, Use Cases, and Market Dynamics

Studying the basic concepts of cloud computing will be extremely beneficial in understanding the move from capital investments in on-premises computing infrastructure to accessing general-purpose computing infrastructure via shared resources over the network on an on-demand basis (Mell and Grance, 2011).

Knowledge about the basics of cloud computing will also help in understanding how different architectural abstractions result in different cloud computing services such as Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS) (Mell and Grance, 2011).

The main differentiation across these three categories is how management responsibilities are shared between the customer and the provider (Mell and Grance, 2011; Alfaujianto et al., 2026).

IaaS provides basic computing, storage, and network services, providing customers with a high degree of control and flexibility in choosing their operating system and runtime environment, without the burden of maintaining physical infrastructure (Alfaujianto et al., 2026).

An example scenario is large-scale, dynamic workloads. For example, Netflix runs on Amazon Web Services (AWS) to stream video content globally (Fortune Business Insights, 2026).

Recently, demand for IaaS has increased due to the need to perform intensive computing tasks for training generative AI models on scalable GPU clusters (Fortune Business Insights, 2026).

The platform-as-a-service (PaaS) model abstracts away the operating system, middleware, and database. PaaS offers a set of preconfigured environments or runtimes that let developers quickly build, test, and deploy new applications without managing the underlying infrastructure (Alfaujianto et al., 2026).

For example, PaaS can accommodate modern software development paradigms such as containerised microservices or continuous integration and deployment pipelines (Fortune Business Insights, 2026).

On the other hand, SaaS provides complete software applications for end users that are delivered over the web (Mell and Grance, 2011).

SaaS provides the lowest cost and quickest adoption, but minimal administrative oversight (Alfaujianto et al., 2026).

SaaS still makes up the majority of software consumption over time, and recent reports show that market leaders are offering self-managing artificial intelligence (AI) agents alongside enterprise resource planning applications (Fortune Business Insights, 2026).

Finally, understanding these delivery methods shows how companies can be most flexible by deliberately matching their requirements to the right amount of control, speed of development, and cost (Alfaujianto et al., 2026).

References

- *Alfaujianto, M., Nugraha, F.R., Muttaqi, F. and Zogara, L.U. (2026) 'Comparative Analysis of Cloud Service Models for Professional Use: IaaS, PaaS, and SaaS', Jurnal Ilmiah Sistem Informasi, 11(1), pp. 1–10.*

- *Fortune Business Insights (2026) Cloud Computing Market Size, Share & Industry Analysis, By Type, By Service (IaaS, PaaS, SaaS), and Regional Forecast, 2026-2034. Report ID: FBI102697. Pune: Fortune Business Insights.*
- *Mell, P. and Grance, T. (2011) The NIST Definition of Cloud Computing. Special Publication 800-145. Gaithersburg: National Institute of Standards and Technology (NIST).*